

CRACKS OF MEANING: CRITICAL WORK WITH AI IN ART EDUCATION

Methodological and Theoretical Guide to a Workshop Model Based on the CAPHE Pilot

This guide grows out of the pilot workshop “Cracks of Meaning: Videopoetry in the Age of AI,” carried out in Kraków from 6–20 March 2026 as part of the International AI Art Biennale. Creative Dialogue and Education with AI. The workshop formed part of the research and educational activities developed within CAPHE – Communities and Artistic Participation in Hybrid Environments, a Horizon Europe project exploring artistic, educational, and social participation across hybrid environments that combine physical and virtual space. The pilot involved the Polish Society of Aesthetics, Blue Point Art London, Apteka Designu, and the Faculty of Interdisciplinary Studies at SWPS University in Kraków. The workshop team included Justyna Gorzkowicz, Aleksandra Łukaszewicz, Jarosław Solecki, and Joanna Szczepanik.

The guide responds to the need to connect art education with critical reflection on artificial intelligence. Its central premise is that new technologies should not be treated only as tools of efficiency, but also as spaces of experiment, uncertainty, and creative tension. Participants are not simply invited to use AI; they are encouraged to observe closely how it processes image, language, and sound, how it shifts meaning, and how it produces new relations between media. The final artistic form proposed in this model is the videopoem—understood not as an illustration of a text, but as a work in which text, image, and sound co-constitute the main structure of meaning.

This approach emerged alongside the work of the international research-art collective Virtually Embodied Critical Thinking – AI (VECT-AI), initiated by Joanna Szczepanik within CAPHE and developed with Justyna Gorzkowicz, Aleksandra Łukaszewicz, Jarosław Solecki, and Joanna Szczepanik. The collective operates in two interconnected fields: first, the theoretical rethinking of critical thinking under conditions of human–AI collaboration; second, practice-based artistic experimentation in XR environments, focused on perception, interpretation, and algorithmic mediation. The workshop model presented here is one result of that convergence.

Rather than offering a closed script to be copied exactly, this guide proposes an open, modular model that can be adapted to different institutions, audiences, and levels of technical experience. At its core is the idea that error, malfunction, and ambiguity are not problems to be removed but productive triggers for reflection and creation.

Audience and Educational Aims

The workshop can be implemented in a range of artistic, cultural, and humanities-based contexts: secondary schools, universities, cultural institutions, research-art projects, and informal learning settings. It is intended primarily for participants aged 16 and above: students, educators, artists, cultural animators, and anyone interested in the relationship between art, technology, and critical media awareness.

In its basic form, the workshop does not require prior experience with AI tools, audiovisual editing, or 3D scanning. Basic digital literacy—using a smartphone, computer, and simple file-sharing platforms—is sufficient, along with willingness to collaborate and experiment. Participants work in small groups, ideally 2–4 people, with internet access, mobile devices, and at least one central computer station.

The workshop's main educational goal is to develop competencies necessary for critical, reflective, and creative engagement with AI in art education. More specifically, it aims to:

- * foster critical thinking in the age of AI, treating AI not as a neutral source of answers but as a system that demands distance, interpretation, and analysis;
- * strengthen cognitive resilience, especially the ability to tolerate uncertainty and resist the seductive fluency and false authority of AI-generated language;
- * expose the imitation of understanding, showing that AI often simulates coherence rather than genuinely comprehending what it “interprets”;
- * develop media and material literacy through artistic practice in hybrid environments, where image, sound, text, and spatial form are continuously transformed;
- * enable participants to assess critically the role of AI in shaping meaning and artistic outcomes.

As a result, participants learn to identify mechanisms of apparent coherence in AI outputs, treat hallucination and error as elements of both artistic and cognitive inquiry, collaborate in hybrid creative processes, and work consciously with the tension between text, image, and sound.

Methodology: Error as Material

The methodological foundation of the workshop lies in reversing conventional habits in both education and artistic production. Instead of pursuing correct, seamless execution, the process is built around malfunction, distortion, and technological limitation.

A key theoretical frame is Productive Failure, developed by Manu Kapur, according to which attempts, struggle, and even failure can generate deeper understanding when properly structured. In this context, failure is not a system defect but a resource that reveals gaps in understanding and activates inquiry. This is especially relevant in art education, where uncertainty and friction are often essential to meaningful creation.

Large language models usually aim to provide immediate, coherent answers. In educational settings, this may reduce the learner's cognitive effort. The workshop therefore deliberately sustains tension by postponing closure and refusing the quick “correct” answer. Participants are encouraged to experience uncertainty rather than eliminate it.

This logic is further reinforced by the workshop's golden rule: do not correct the error. The principle draws on Jarosław Solecki's concept of the “dystopia of imitation,” which argues that the technological drive toward perfect imitation—such as a flawless 1:1 scan of reality—is inherently fragile and dystopian. Breakdowns, glitches, and deformations become moments of epistemic revelation. Participants do

not simply “catch the model making mistakes”; they learn to recognize the moment when they themselves begin to trust a smooth answer merely because it sounds plausible.

In this model, imperfect 3D scans—with holes, blurs, and impossible fusions of form—are crucial. They reveal that the algorithm does not understand space or image; it only interpolates and averages data. AI-generated interpretations of such images then become evidence of how machines simulate meaning rather than comprehend it.

The Videopoem as Critical Form

The workshop takes the videopoem as its recommended output form. This choice is grounded in reflections by Tom Konyves and Ian Cottage, who, from different angles, emphasize the specific relation of text, moving image, and sound. In this guide, the videopoem is understood as a nonlinear form in which these elements remain in tension and jointly produce meaning. It is neither a simple visualization of a finished poem nor a free audiovisual piece loosely borrowing poetic qualities.

This form has strong pedagogical value. By rejecting the demand for linear storytelling, it frees participants from the pressure of conventional narrative and directs their attention to rhythm, pause, juxtaposition, fragmentation, and error. This is especially important in critical work with AI, since AI systems tend to produce smooth, narrative closure. The videopoem resists that impulse by preserving ambiguity and multiplicity.

Workshop Structure

The workshop is structured in three stages.

Day 1: Error as Material / AI and the Closure of Meaning

Participants generate imperfect digital copies of physical objects using a mobile 3D scanning app such as KIRI Engine. They deliberately record poor-quality footage—moving too fast, changing angles abruptly, producing incomplete scans. These materials form the initial visual archive: video files, 3D models, and screenshots.

The groups then select not the “best” images, but those in which deformation, noise, and ambiguity are most pronounced. These screenshots are uploaded to a language model such as ChatGPT, which is asked to describe, interpret, or narrativize the image using simple, repeatable prompts like: “Provide an emotional interpretation of this image” or “Tell the story suggested by this image.” Participants do not correct hallucinations or misreadings. Instead, they collect phrases, fragments, and verbal excesses generated by the system. These form the Text Bank for later composition.

Day 2: Composition of Text, Sound, and Image

The second day moves from collection to composition. Participants work in specialist groups focused on the textual, sound, and, in an extended version, also the visual layer. The textual group develops a shared verbal layer from the AI-generated

material, deciding whether text will function as caption, voice, rhythm, movement, or silence. The sound group creates an auditory landscape. In the expanded version, participants also continue shaping the visual material more independently.

The process is coordinated through the central workstation known as CENTRALA, which manages all files and supports live decision-making. All group materials feed into a shared structure called the VIDEPOEM ATM, a scalable common archive containing video, 3D models, screenshots, and texts. From this archive, the final composition is assembled.

The crucial moment is the performative edit: montage is not treated as hidden postproduction, but as a collective live process in which decisions about sequence, rhythm, tension, text, and sound are negotiated in real time between participants, facilitators, and the operator of CENTRALA.

Day 3: Presentation and Meta-Reflection

The final stage is a public or internal presentation of the videopoem, ideally as a multi-screen installation with a shared sound layer. Such a format preserves the nonlinear structure of the work and requires viewers to shift attention between image, text, and sound rather than follow a single linear path.

This stage is followed by moderated discussion and meta-reflection. Participants consider how AI attempted to close meaning, where human agency remained decisive, and how authorship should be understood in such hybrid processes. Questions such as “To what extent is AI the author of the final work?” or “Did you feel in control of the tool and the outcome?” help consolidate the workshop as both an artistic and a critical-thinking experience.

Tools, Ethics, and Implementation

The workshop relies on an integrated ecosystem of tools: mobile devices for scanning and recording, a language model with image analysis capacity, a cloud-based file-sharing environment, and software for audio and video editing. Google Drive functions as the infrastructural backbone of CENTRALA; tools such as Audacity, text-to-speech software, and nonlinear video editing software support the final composition. The synthetic voice is conceptually justified within the model, since a neutral or mechanical reading can intensify the effect of AI’s false descriptive authority.

The reflective dimension of the workshop is inseparable from its ethical framework. Participants should always be informed when and how AI is used. AI-generated fragments should be clearly marked. Facts and outputs should be critically verified rather than accepted as truth. Consent must be obtained for the use of participants’ images or materials, and authorship arrangements should be transparent. A short discussion of ethical issues—drawing, for example, on UNESCO guidance for generative AI in education and research—is recommended at the beginning of the workshop.

For implementation, institutions should provide stable Wi-Fi, projection equipment, sound, tables for group work, charging access, and a central computer connected to the shared cloud environment. The model works best with 8–16 participants, though it can scale up to about 20 if work is carefully distributed. A facilitation team of 3–4 people is recommended, combining expertise in the humanities, pedagogy, art, and digital technologies.

An important extension of the workshop is its archival and exhibition dimension. The final videopoem may be presented as a multi-screen installation, while the process and results can also be documented in a 3D virtual environment accessible through a web browser. Such documentation should not hide distortions and glitches; instead, it should preserve them as traces of process and as carriers of meaning. In the pilot, this documentation was developed as a web application in JavaScript (React) using Three.js, based on scans created with KIRI Engine and Blender, and now forms part of the Blue Point Art Gallery archive.

Conclusion

“Cracks of Meaning” proposes a model of art education in which AI is neither celebrated as a transparent creative partner nor rejected as a threat, but approached as an unstable and revealing system. By working with error, ambiguity, and algorithmic overconfidence, participants learn not only to create, but to think critically with and against the technologies that increasingly shape cultural production. In this sense, the workshop is both a method of artistic making and a practice of epistemic resistance.